

Recuperare informazioni per le meta-analisi

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recuperare informazioni descrittive



① media: 49 anni

② s.d.: 27.2 anni

mediana ?
quartili ?

oppure: mediana 44, primo quartile 41, terzo quartile 74.

Media? Deviazione standard?

recuperare informazioni descrittive / 1 of 2

Possiamo dedurre la media μ dalla mediana, dai quartili e dal range?

$$\mu \approx \frac{a + 2m + b}{4} + \frac{a - 2m + b}{4n} \approx \frac{a + 2m + b}{4}$$

$$\mu \approx \frac{a + 2Q_1 + 2m + 2Q_3 + b}{8}$$

$$\mu \approx \frac{Q_1 + m + Q_3}{3}$$

recuperare informazioni descrittive / 2 of 2

Possiamo dedurre la deviazione standard σ dalla mediana, dai quartili e dal range?

$$\sigma \approx \frac{b - a}{\xi(n)}$$

$$\sigma \approx \frac{Q_3 - Q_1}{\eta(n)}$$

$$\sigma \approx \frac{1}{2} \left(\frac{b - a}{\xi(n)} + \frac{Q_3 - Q_1}{\eta(n)} \right)$$

Un esempio



- the Reference Data Set
 - dal 2003 al 2014, King's College London Dental Institute
- LL8Gf
 - Lower Left Third Molar
 - Stage G female
 - 18 years old
 - caucasian UK

Un esempio

<https://www.dentalage.co.uk/rds-uk-caucasian/>

$$n = 114$$

$$\mu \approx \frac{a + 2Q_1 + 2m + 2Q_3 + b}{8}$$

$$\frac{14.9 + 34.2 + 36.2 + 39.9 + 22.8}{8}$$

$$\mu \approx 18.5$$

0th%ile	0.5%ile	5%ile	10%ile
14.96	15.73	15.97	16.32
25%ile	50%ile	75%ile	90%ile
17.07	18.13	18.99	20.64
95%ile	99.5%ile	100th%ile	
21.35	21.91	22.78	

Un esempio

<https://www.dentalage.co.uk/rds-uk-caucasian/>

$$n = 114$$

$$\sigma \approx \frac{1}{2} \left(\frac{b - a}{\xi(n)} + \frac{Q_3 - Q_1}{\eta(n)} \right)$$

$$\frac{1}{2} \left(\frac{22.8 - 14.9}{4.55} + \frac{19.0 - 17.1}{1.35} \right)$$

$$\sigma \approx 1.6$$

0th%ile	0.5%ile	5%ile	10%ile
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14.96	15.73	15.97	16.32
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25%ile	50%ile	75%ile	90%ile
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17.07	18.13	18.99	20.64
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95%ile	99.5%ile	100th%ile
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21.35	21.91	22.78
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Un esempio

$$\mu \approx 18.5$$

$$\sigma \approx 1.6$$

TDS	n	mean	sd
LL8Gf	114	18.25	1.63

Commentary to Wan et al. (2014):
Estimating the standard deviation
from the sample size and range or quartiles

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Abstract

This short note proposes two additive corrections to a pair of relations published in Wan et al. [WWLT14] in order to extend them to a 'small sample size' condition. In particular we focus the interest on the possibility to provide an estimate to the sample standard deviation σ when knowing only the sample size n , the range $[a, b]$ and/or the quartiles Q_1, Q_3 of some data. Our results allow to explicitly compute σ , for instance with software R or any spreadsheet, for any sample size $n \geq 2$. R codes and data are publicly available on <https://github.com/MassimoBorelli/sd>

